

Hybrid Linear-Convex Confidence: An Asymmetric Scoring Function for Language Model Calibration

Dr Simon McCallum
Victoria University of Wellington
`simon.mccallum@vuw.ac.nz`

September 12, 2026

Abstract

When a student answers a multiple-choice question, the cost of a confident wrong answer is qualitatively different from the cost of a cautious correct one. Yet the standard tools for measuring calibrated confidence, notably the Brier score and log-loss, treat over- and under-confidence symmetrically. Large language models inherit this blind spot. Trained on cross-entropy loss, they are systematically overconfident, with no penalty structure that distinguishes dangerous certainty from harmless caution. We propose a Hybrid Linear-Convex Confidence (HLCC) scoring function, which pairs a linear reward $R(c) = 1+c$ for correct answers with a quadratic penalty $P(c) = -2c^2$ for confident errors. The resulting optimal confidence $c^* = p/[4(1-p)]$ is conservative: strictly below the true accuracy p for $p < 0.75$, matching it at $p = 0.75$, and saturating at $c^* = 1$ when $p \geq 0.80$. Removing the upper bound on c yields an *unbounded wagering* extension where $c^* \approx 8$ at $p = 0.97$, but we show that even modest uncertainty in the estimate of p itself, arising from statistical noise, question ambiguity, and historical overconfidence, makes such aggressive staking catastrophically fragile, providing a principled argument for conservative confidence. We benchmark HLCC alongside 13 baseline confidence methods across 5 models (7B–14B parameters) and 9 datasets, evaluating discrimination (AUROC), calibration (ECE, Brier), selective prediction (PRR), and HLCC score. As a case study, we train a lightweight *NormShift* confidence head that exploits per-layer normalisation dynamics, achieving near-perfect in-domain discrimination on some architectures but limited out-of-distribution transfer, underscoring that learned confidence heads complement rather than replace logit-based methods.

1 Introduction

Consider a student sitting a multiple-choice examination. She reads the question, eliminates one implausible option, and is left with three candidates, one of which she is fairly sure is correct. How confident should she be? If the exam awards one mark for a correct answer and deducts nothing for an incorrect one, the rational strategy is to always answer, even when guessing. If instead the exam deducts a full mark for every wrong answer, a risk-averse student will leave the question blank, forfeiting partial knowledge she genuinely possesses. Neither scheme encourages her to *think carefully about how much she knows*.

This poses a dilemma regarding the design of scoring systems. We want a student, or a language model, to participate rather than abstain, yet we also want confident errors to be costly enough that the agent invests effort in calibrating its own uncertainty. The classical solution is the *proper scoring rule*: the Brier score and logarithmic loss are both maximised when the reported confidence equals the true probability of being correct. However, this mathematical elegance creates a symmetric treatment of over- and under-confidence. A model that reports 90% confidence when it is right 80% of the time incurs the same penalty as one that reports 70%

confidence at the same accuracy. In medicine, law, and autonomous systems, the consequences of a confidently wrong answer are qualitatively different from those of a cautiously correct one.

Gardner-Medwin’s certainty-based marking (CBM) for educational assessment [10, 11, 30] recognised this asymmetry decades ago, encoding it through discrete penalty scales derived from logarithmic proper scoring rules [17]. Under CBM, a student who marks high confidence and is wrong loses substantially more than one who hedges. The key insight is simple: in most real-world settings, it is better to err on the side of under-confidence than over-confidence. A doctor who says “I’m not sure, let me check” causes far less harm than one who confidently prescribes the wrong treatment.

Inspired by this asymmetry, we propose the Hybrid Linear-Convex Confidence (HLCC) scoring function:

$$S(y, c) = \begin{cases} 1 + c & \text{if } y = 1 \text{ (correct),} \\ -2c^2 & \text{if } y = 0 \text{ (incorrect),} \end{cases} \quad (1)$$

where $c \geq 0$ is the stated confidence and $y \in \{0, 1\}$ indicates correctness. The linear reward provides a constant marginal incentive for increased confidence on correct predictions, while the quadratic penalty creates an escalating marginal cost for confident errors. At $c = 0$, a correct answer still earns one point and an incorrect answer costs nothing, so participating is always dominant over abstention and our student never has reason to leave a question blank.

In the *bounded* variant ($c \in [0, 1]$), this structure naturally induces conservative confidence estimates: the optimal c^* lies strictly below the true accuracy p for $p < 0.75$. In the *unbounded* variant ($c \in [0, \infty)$), it becomes a wagering system where a model can stake arbitrarily high confidence, earning large rewards when correct but suffering catastrophic quadratic penalties when wrong. There is a further subtlety in the unbounded regime: the optimal stake assumes the model knows its own accuracy perfectly, yet in practice there is always uncertainty in the estimate of p itself. If the true accuracy is even a few percent lower than believed, say 95% rather than 97%, the error rate nearly doubles and the quadratic penalty amplifies that shortfall dramatically. This uncertainty-in- p effect provides an additional argument for the conservative bias that HLCC naturally encodes.

Large language models trained on internet text are systematically overconfident [18, 19], and the standard tools for calibration treat this overconfidence no differently from equivalent under-confidence. A scoring function should reflect the real-world cost structure: confident correctness is valuable, confident error is dangerous, and cautious uncertainty is safe. HLCC encodes precisely this structure.

Contributions.

1. We derive the theoretical properties of HLCC, showing that the optimal confidence $c^* = p/[4(1-p)]$ is strictly below the true accuracy p for $p < 0.75$, and we analyse the unbounded extension where $c^* > 1$ creates a wagering system with expected payoff $p(8-7p)/[8(1-p)]$. We further show that uncertainty in the estimate of p strengthens the case for conservative confidence (Section 2).
2. We conduct a comprehensive benchmarking study of 14 confidence methods across 5 models (7B–14B), 9 datasets, and 6 metric families, using HLCC score alongside standard metrics to reveal overconfidence that ECE and AUROC miss (Sections 4 and 6).
3. As a case study, we propose the *NormShift* confidence head, a lightweight architecture trained with HLCC loss that exploits per-layer normalisation dynamics (Section 3).

2 Theoretical Framework

2.1 The HLCC Scoring Function

Definition 1 (HLCC Score). *For a binary outcome $y \in \{0, 1\}$ and stated confidence $c \geq 0$, the HLCC score is*

$$S(y, c) = y(1 + c) + (1 - y)(-2c^2). \quad (2)$$

We distinguish two variants: bounded HLCC with $c \in [0, 1]$, and unbounded HLCC with $c \in [0, \infty)$.

The reward function $R(c) = 1 + c$ is affine with constant derivative $R'(c) = 1$. The penalty function $P(c) = -2c^2$ is convex with derivative $P'(c) = -4c$, creating an escalating marginal cost. At $c = 0$, the score is $S(1, 0) = 1$ and $S(0, 0) = 0$, establishing a *safe guess* baseline: answering at zero confidence is always weakly dominant over abstention.

2.2 Optimal Confidence

Proposition 1 (Optimal Confidence under HLCC). *For an agent with true probability of correctness $p \in (0, 1)$, the confidence c^* maximising expected HLCC score is*

$$c^* = \frac{p}{4(1 - p)}. \quad (3)$$

Under bounded HLCC ($c \in [0, 1]$), this is clamped: $c_{\text{bounded}}^ = \min(1, p/[4(1 - p)])$.*

Proof. The expected score as a function of stated confidence c is

$$\mathbb{E}[S] = p(1 + c) + (1 - p)(-2c^2) = p + pc - 2(1 - p)c^2. \quad (4)$$

Differentiating with respect to c :

$$\frac{\partial \mathbb{E}[S]}{\partial c} = p - 4(1 - p)c. \quad (5)$$

Setting to zero and solving for c :

$$c^* = \frac{p}{4(1 - p)}. \quad (6)$$

The second derivative $\partial^2 \mathbb{E}[S]/\partial c^2 = -4(1 - p) < 0$ for $p < 1$, confirming this is a global maximum on $[0, \infty)$. Under the bounded constraint $c \in [0, 1]$, we take $\min(1, c^*)$. $\square$

Corollary 1 (Crossover and Saturation). *The optimal confidence equals the true accuracy ($c^* = p$) at exactly $p = 3/4$. For $p < 3/4$, $c^* < p$ (conservative). For $p > 3/4$, $c^* > p$. Under bounded HLCC, saturation occurs at $c^* = 1$ when $p \geq 4/5$. Under unbounded HLCC, c^* grows without limit: $c^* = 2.25$ at $p = 0.9$, $c^* \approx 8$ at $p = 0.97$, and $c^* \rightarrow \infty$ as $p \rightarrow 1$.*

Proof. Setting $c^* = p$: $p = p/[4(1 - p)]$ implies $4(1 - p) = 1$, hence $p = 3/4$. The bounded saturation point solves $p/[4(1 - p)] = 1$, giving $p = 4/5$. As $p \rightarrow 1$, the denominator $4(1 - p) \rightarrow 0$, so $c^* \rightarrow \infty$. $\square$

Proposition 2 (Maximum HLCC — Bounded). *Under bounded HLCC ($c \in [0, 1]$), the maximum achievable expected score for a model with accuracy p is $S_{\text{max}} = 2p$, attained when $c = 1$ for every correct prediction and $c = 0$ for every incorrect prediction. The bounded HLCC efficiency $\eta = S_{\text{HLCC}}/(2p)$ measures how well the confidence method exploits the model’s accuracy.*

Proof. With perfect self-knowledge, each correct prediction (p fraction) scores $S(1, 1) = 1 + 1 = 2$, and each incorrect prediction ($1 - p$ fraction) scores $S(0, 0) = 0$. The expected score is $p \cdot 2 + (1 - p) \cdot 0 = 2p$. No other bounded confidence assignment achieves a higher expected score, since $R(c) \leq 2$ and $P(c) \leq 0$ on $[0, 1]$. $\square$

2.3 Unbounded Confidence: The Wagering Extension

Removing the upper bound on c transforms HLCC from a calibration score into a *wagering system*. A model that is highly confident can stake $c \gg 1$, earning a large linear reward when correct but suffering a catastrophic quadratic penalty when wrong.

Proposition 3 (Optimal Expected Score — Unbounded). *Under unbounded HLCC, a model with accuracy $p \in (0, 1)$ that plays the optimal confidence $c^* = p/[4(1-p)]$ achieves expected score*

$$\mathbb{E}[S(c^*)] = \frac{p(8-7p)}{8(1-p)}. \quad (7)$$

This is finite for all $p < 1$ and grows without bound as $p \rightarrow 1$.

Proof. Substituting $c^* = p/[4(1-p)]$ into $\mathbb{E}[S] = p + pc^* - 2(1-p)c^{*2}$:

$$pc^* = \frac{p^2}{4(1-p)}, \quad (8)$$

$$2(1-p)c^{*2} = 2(1-p) \cdot \frac{p^2}{16(1-p)^2} = \frac{p^2}{8(1-p)}. \quad (9)$$

$$\text{Hence } \mathbb{E}[S(c^*)] = p + \frac{p^2}{4(1-p)} - \frac{p^2}{8(1-p)} = p + \frac{p^2}{8(1-p)} = \frac{8p(1-p)+p^2}{8(1-p)} = \frac{p(8-7p)}{8(1-p)}. \quad \square$$

Proposition 4 (Break-Even Confidence). *For a given confidence level c , the minimum accuracy required for positive expected HLCC score is*

$$p_{\min}(c) = \frac{2c^2}{1+c+2c^2}. \quad (10)$$

The confidence c is optimal (better than $c-1$) only when $p > 4c/[1+4c]$.

Proof. Setting $\mathbb{E}[S] = p(1+c) - 2(1-p)c^2 = 0$ and solving for p : $p(1+c+2c^2) = 2c^2$, giving (10). $\square$

Remark 1 (Worked example: $c = 9$). *At $c = 9$, each correct answer scores $R(9) = 10$ and each error costs $P(9) = -162$. The break-even accuracy is $p_{\min} = 162/172 \approx 0.942$. Setting $c = 9$ is optimal over $c = 8$ when $p > 34/35 \approx 0.971$, since the marginal gain from staking one additional unit of confidence requires accuracy sufficient to offset the quadratic penalty increase from -128 to -162 .*

The unbounded extension connects HLCC to classical wagering theory. The quadratic penalty plays the role of a *risk charge* analogous to the house edge in gambling [23]: the linear reward makes confident correct answers “cheap,” while the quadratic penalty makes confident errors “expensive.” This asymmetry is precisely the structure that Gardner-Medwin’s CBM discretises into three levels [11]: CBM level $C=3$ (+3/−6) corresponds roughly to $c \approx 1.5$ in unbounded HLCC, and the CBM boundary at $p = 4/5$ matches the HLCC saturation point where $c^* = 1$.

Table 1 shows the optimal confidence mapping at representative accuracy levels for both bounded and unbounded variants.

2.4 Uncertainty in the Estimate of Accuracy

The analysis above assumes the agent knows its own accuracy p perfectly. In practice, no system, whether human or artificial, has exact self-knowledge. A student who believes she answers correctly 97% of the time is in reality correct only 95% of the time. A language model’s empirical accuracy on a validation set is an estimate subject to sampling variance, distribution shift, and the inherent difficulty of self-assessment. This uncertainty in p has surprisingly large consequences in the unbounded regime.

Table 1: Optimal confidence c^* under bounded and unbounded HLCC, compared with Brier ($c^* = p$). The unbounded expected score $\mathbb{E}[S(c^*)]$ grows rapidly for high p , rewarding models that can reliably identify near-certain answers.

p	Bounded c^*	Unbounded c^*	Brier c^*	$\mathbb{E}[S(c^*)]$	Regime
0.00	0.000	0.000	0.00	0.000	Abstention
0.25	0.083	0.083	0.25	0.260	Suppressed guessing
0.50	0.250	0.250	0.50	0.563	Conservative
0.75	0.750	0.750	0.75	0.891	Crossover ($c^* = p$)
0.80	1.000	1.000	0.80	1.000	Bounded saturation
0.85	1.000	1.417	0.85	1.176	<i>Unbounded: $c > 1$</i>
0.90	1.000	2.250	0.90	1.913	
0.93	1.000	3.321	0.93	2.953	
0.95	1.000	4.750	0.95	4.156	
0.97	1.000	8.083	0.97	7.331	
0.99	1.000	24.750	0.99	24.009	

Proposition 5 (Cost of Accuracy Overestimation). *Suppose an agent believes its accuracy is $\hat{p}$ and plays the corresponding optimal confidence $\hat{c} = \hat{p}/[4(1-\hat{p})]$, but the true accuracy is $p = \hat{p} - \varepsilon$ for some $\varepsilon > 0$. The expected score under the true accuracy is*

$$\mathbb{E}[S(\hat{c})] = p(1 + \hat{c}) - 2(1-p)\hat{c}^2. \quad (11)$$

The shortfall relative to perfect self-knowledge grows quadratically in $\hat{c}$ and hence explodes as $\hat{p} \rightarrow 1$.

To see why this matters, consider a concrete example. An agent believes $\hat{p} = 0.97$ and accordingly stakes $\hat{c} = 0.97/[4 \times 0.03] \approx 8.08$. Each correct answer earns $R(8.08) = 9.08$, and each error costs $P(8.08) = -2 \times 8.08^2 \approx -130.7$. If the true accuracy is in fact $p = 0.95$, a mere two percentage points lower, the error rate rises from 3% to 5%, an increase of nearly 67%. The expected score drops from 4.89 (under correct self-knowledge at $p = 0.97$) to $0.95 \times 9.08 - 0.05 \times 130.7 \approx 2.10$. Had the agent known its true accuracy and staked optimally at $\hat{c} = 0.95/[4 \times 0.05] = 4.75$, its expected score would have been 3.21, more than 50% higher.

The mechanism is straightforward. In the unbounded regime, the optimal confidence grows as $p/[4(1-p)]$, which is extremely sensitive to the denominator. A small overestimate of p produces a large overestimate of c^* , and the quadratic penalty amplifies the resulting errors. There are two reasons that this sensitivity matters for practical system design. The first is that it provides a principled argument for conservative confidence even in the unbounded regime: an agent that is uncertain about its own accuracy should stake less aggressively than the theoretical optimum. The second is that it highlights the value of the bounded variant for deployment, where clamping $c \leq 1$ limits the maximum possible loss to -2 per question regardless of how badly accuracy is overestimated.

Table 2 illustrates this effect across several accuracy levels with a fixed overestimation of $\varepsilon = 0.02$. At moderate accuracy the cost is modest, less than 1% loss at $\hat{p} = 0.82$. However, in the high-accuracy regime the cost becomes catastrophic: at $\hat{p} = 0.99$ with true $p = 0.97$, the agent stakes $c \approx 24.75$ and the expected score turns *negative*, meaning the agent would have been better off not answering at all. It is precisely the fact that real-world accuracy estimates are never perfectly reliable that makes the unbounded regime dangerous without additional safeguards.

Sources of second-order uncertainty. The simple ε -offset model above understates the problem, because it treats the estimation error as a single scalar. In practice, the gap between

Table 2: Effect of accuracy overestimation in unbounded HLCC. The agent believes $\hat{p}$ and stakes optimally for that belief, but the true accuracy is $p = \hat{p} - 0.02$. The *loss ratio* is the ratio of actual expected score to the score achievable under correct self-knowledge at the true p .

$\hat{p}$	True p	$\hat{c}$	True optimal c^*	$\mathbb{E}[S(\hat{c})]$	Loss ratio
0.82	0.80	1.139	1.000	1.192	0.994
0.87	0.85	1.673	1.417	1.432	0.986
0.92	0.90	2.875	2.250	1.834	0.959
0.95	0.93	4.750	3.321	2.189	0.885
0.97	0.95	8.083	4.750	2.095	0.653
0.99	0.97	24.750	8.083	-11.78	< 0

believed and true accuracy arises from at least three distinct sources, each of which compounds the others.

The first is *statistical model noise*: the base variance inherent in any finite evaluation. A model tested on 300 questions has a 95% confidence interval of roughly $\pm 3\%$ around its observed accuracy. This is irreducible sampling error, approximately constant regardless of the accuracy level.

The second is *question ambiguity*, or setter risk. The agent may be solving the wrong problem entirely. The question may be poorly phrased, contain a semantic trick, or rely on a convention the agent does not share. This introduces a noise floor that does not disappear simply because the event itself is likely. A model that is “certain” it knows the capital of Myanmar may be confidently answering a different question than the one being asked.

The third, and most consequential for the unbounded regime, is the *epistemic humility penalty*: the historical rate at which knowledge the agent believed to be true has turned out to be false. Unlike the first two sources, this penalty *grows* as claimed confidence approaches certainty. A well-calibrated agent should recognise that its track record of “absolute certainty” includes a non-trivial error rate. This metacognitive risk, the risk of being wrong about being right, balloons precisely in the regime where the unbounded formula prescribes the most aggressive stakes.

When these three sources are compounded, the effective uncertainty band never approaches zero, even at $\hat{p} \rightarrow 1$. The practical consequence is that the optimal bet c^* should remain capped at a finite, conservative level rather than growing without bound. This is precisely the behaviour that bounded HLCC enforces by construction, and it suggests that the bounded variant is not merely a mathematical convenience but a reflection of the epistemic reality that no agent, human or artificial, ever truly has zero probability of error.

2.5 Comparison with Proper Scoring Rules

HLCC is *not* a strictly proper scoring rule. Unlike the Brier score $S_B = -(c - y)^2$ and log-loss $S_L = y \log c + (1 - y) \log(1 - c)$, which are both maximised at $c^* = p$, HLCC imposes a conservative transformation of accuracy to confidence.

Proposition 6 (Reward–Penalty Asymmetry). *At any confidence level $c > 0$, the marginal penalty for an incorrect prediction exceeds the marginal reward for a correct one:*

$$|P'(c)| = 4c > 1 = R'(c) \quad \text{for all } c > 0.25. \quad (12)$$

The marginal risk-to-reward ratio is $4c : 1$.

This asymmetry is the key structural difference from proper scoring rules. At $c = 1$, the risk-to-reward ratio is $4 : 1$ (marginal) or $2 : 1$ (absolute: gaining 1 point from $R(0) = 1$ to

$R(1) = 2$, but risking a 2-point loss from $P(0) = 0$ to $P(1) = -2$). In the unbounded variant, this ratio grows as $4c : 1$ —at $c = 9$, the marginal risk is $36 : 1$, ensuring that only models with very high accuracy ($p > 0.97$) benefit from such aggressive staking.

2.6 Safe Guess Mechanism

At $c = 0$, the score is $S(1, 0) = 1$ (correct) and $S(0, 0) = 0$ (incorrect). The expected value of a random guess ($p = 0.25$ for 4-option MCQ) at $c = 0$ is $0.25 \times 1 = 0.25 > 0$. This ensures that participating is always dominant over abstention, resolving the *omission bias* observed in negative-marking schemes where risk-averse test-takers skip questions they partially know [2].

3 Architecture

3.1 NormShift Confidence Head

We propose the *NormShift* confidence head, a lightweight module trained on top of a frozen language model to predict per-token confidence. The architecture exploits an observation about transformer normalisation layers: the magnitude of the correction applied by LayerNorm encodes information about the model’s internal uncertainty.

Definition 2 (Norm-Shift Signal). *For layer ℓ of a transformer with LayerNorm producing output $\mathbf{h}^{(\ell)}$ from pre-normalisation input $\mathbf{x}^{(\ell)}$, the norm-shift signal is*

$$\delta^{(\ell)} = \|\mathbf{h}^{(\ell)}\|_2 - \|\mathbf{x}^{(\ell)}\|_2, \quad (13)$$

measuring how much the normalisation operation altered the representation magnitude.

Dual-Path Architecture. The NormShift head processes two complementary information streams:

- **Path A** (Norm-Shift Path): The vector $\boldsymbol{\delta} = [\delta^{(1)}, \dots, \delta^{(L)}] \in \mathbb{R}^L$ of per-layer norm-shift signals is projected through a linear layer with GELU activation to an intermediate representation of dimension $d_{\text{mid}} = \max(64, 2L)$.
- **Path B** (Hidden-State Path): The final hidden state $\mathbf{h}^{(L)} \in \mathbb{R}^d$ is projected through a separate linear layer with GELU activation to the same intermediate dimension d_{mid} .

The two paths are concatenated and processed by a combination network:

$$c = \sigma\left(\frac{1}{\tau} \mathbf{w}^\top \text{GELU}\left(\mathbf{W}_{\text{comb}} \begin{bmatrix} f_A(\boldsymbol{\delta}) \\ f_B(\mathbf{h}^{(L)}) \end{bmatrix} + \mathbf{b}_{\text{comb}}\right)\right), \quad (14)$$

where τ is a learnable temperature parameter initialised to 1, and σ is the sigmoid function. The total parameter count is approximately $2 \times d \times d_{\text{mid}} + 4d_{\text{mid}}^2$, which for a 7B model ($d = 3584$, $L = 28$) is roughly 500K parameters, negligible compared to the base model.

Training Objective. The confidence head is trained with the HLCC loss from Equation (1), combined with a cross-entropy warmup:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \mathcal{L}_{\text{HLCC}}, \quad (15)$$

where α decays from 1 to a target value over a warmup period. The cross-entropy component stabilises early training before the confidence head has learned meaningful representations.

3.2 Uncertainty-Aware LayerNorm

To capture the doubt signals, we extend standard LayerNorm to record pre-normalisation statistics:

$$\text{UncertaintyAwareLayerNorm}(\mathbf{x}) = \text{LayerNorm}(\mathbf{x}), \quad \text{storing} \quad \begin{bmatrix} \frac{\|\mathbf{x}\|_2}{\sqrt{\text{Var}(\mathbf{x}) + \epsilon}} \\ \text{Var}(\mathbf{x}) \end{bmatrix} \quad (16)$$

These three signals per layer (pre-norm magnitude, correction scale, and variance) form the input to the norm-shift path. For inference with a pre-trained frozen model, only the norm-shift signals from `output_hidden_states=True` are needed; no modification to the base model is required.

4 Benchmarking Methodology

4.1 Models

We evaluate 9 instruction-tuned models spanning 1.1B to 14.2B parameters. Models are loaded at the highest precision that fits within the 32 GB VRAM budget:

Table 3: Models evaluated with precision used for each scale.

Key	Model	Parameters	Precision
TinyLlama	TinyLlama-1.1B-Chat-v1.0	1.1B	FP16
Qwen2-1.5B	Qwen2-1.5B-Instruct	1.5B	FP16
Phi-2	microsoft/phi-2	2.7B	FP16
Phi-3-mini	Phi-3-mini-4k-instruct	3.8B	FP16
Mistral-7B	Mistral-7B-Instruct-v0.3	7.2B	FP16
Qwen2.5-7B	Qwen2.5-7B-Instruct	7.6B	FP16
Llama-3.1-8B	Llama-3.1-8B-Instruct	8.0B	FP16
Qwen2.5-14B	Qwen2.5-14B-Instruct	14.2B	8-bit
DeepSeek-R1-14B	DeepSeek-R1-Distill-Qwen-14B	14.2B	8-bit

4.2 Datasets

We use 9 multiple-choice QA datasets covering diverse reasoning domains:

Table 4: Evaluation datasets. All converted to uniform MCQ format.

Dataset	Source	Options	Domain
TruthfulQA	[27]	4-5	Calibration / misconceptions
ARC-Easy	[4]	4	Science (elementary)
ARC-Challenge	[4]	4	Science (hard)
MMLU	[20]	4	57 subjects
HellaSwag	[44]	4	Commonsense completion
MMLU-Pro	[39]	10	Extended MMLU (A-J)
GSM8K	[5]	4	Math reasoning (converted)
BoolQ	[3]	2	Yes/no QA
TriviaQA	[21]	4	Open-domain QA (converted)

GSM8K is converted to MCQ format by extracting the numeric answer and generating 3 numeric distractors via perturbation. TriviaQA is converted by sampling 3 distractor answers from other questions in the dataset. BoolQ maps directly to 2-choice MCQ.

4.3 Confidence Methods

We evaluate 14 confidence estimation methods (13 baselines plus our NormShift head) organised in three tiers by computational cost, plus two combination methods.

Tier 1: Single Forward Pass. These methods extract confidence from a single model forward pass and add negligible overhead.

Entropy Shannon entropy of the softmax distribution over answer-choice tokens, normalised to $[0, 1]$.

Top- k Ratio of top-1 to top-2 logit probabilities.

MSP Maximum softmax probability over choice tokens [19].

G-NLL Greedy negative log-likelihood of the selected token.

NormShift (HLCC) Trained dual-path confidence head using norm-shift signals and HLCC loss (ours).

Tier 2: Post-Hoc Recalibration. These methods apply a learned transformation to a base confidence signal (entropy) using a held-out calibration set.

Temperature Scaling Single learned temperature parameter T [18].

Isotonic Regression Non-parametric monotonic recalibration using the pool-adjacent-violators algorithm [43].

Platt Scaling Two-parameter logistic regression $\sigma(a \cdot \text{logit}(c) + b)$ [33].

Tier 3: Multi-Pass Generation. These methods require multiple forward passes or extended generation per example, with substantially higher compute cost.

Verbalized Prompt the model to self-report confidence as a number.

Self-Consistency Sample $n = 10$ answers and take the majority vote frequency as confidence [38].

Semantic Entropy Cluster n sampled answers by semantic equivalence and compute Shannon entropy over cluster probabilities [24].

CoT Confidence Two-stage prompting: generate chain-of-thought reasoning, then extract answer and confidence from a structured JSON response.

Additionally, we evaluate two combination methods:

LVU Linguistic Verification of Uncertainty: generate an explanation and detect hedging language (“might”, “perhaps”) vs. certainty language (“definitely”, “clearly”).

CoCoA Confidence Combination Approach: weighted average of entropy and self-consistency signals.

4.4 Evaluation Metrics

We report metrics across three complementary axes.

Discrimination. Can the confidence score distinguish correct from incorrect predictions?

- **AUROC:** Area under the ROC curve, treating confidence $> t$ as a binary classifier for correctness.
- **AUPRC:** Area under the precision-recall curve, more informative than AUROC when the class distribution is imbalanced [34].

Calibration. Does the confidence match the empirical frequency of correctness?

- **ECE:** Expected Calibration Error with 10 equal-width bins [31].
- **Brier Score:** Mean squared error between confidence and binary correctness, $\frac{1}{n} \sum (c_i - y_i)^2$.
- **Overconfidence Rate:** Fraction of incorrect predictions with confidence > 0.7 .

Selective Prediction. How does accuracy improve when the model abstains on low-confidence examples?

- **PRR:** Prediction Rejection Ratio. Measures the area between the model’s accuracy-vs-rejection curve and the random baseline, normalised by the oracle upper bound [29]:

$$\text{PRR} = \frac{\text{AULC}_{\text{random}} - \text{AULC}_{\text{model}}}{\text{AULC}_{\text{random}} - \text{AULC}_{\text{oracle}}}. \quad (17)$$

- **Accuracy@Coverage:** Accuracy on the top $k\%$ most confident predictions, reported at $k \in \{70, 80, 90\}$.

HLCC-Specific.

- **HLCC Score:** Mean score under Equation (1), measuring confidence-weighted performance.
- **Theoretical Max HLCC:** The maximum achievable HLCC score for a model with accuracy p is $S_{\text{max}} = 2p$, attained when the model reports $c = 1$ on every correct prediction and $c = 0$ on every incorrect prediction (perfect knowledge of its own correctness).
- **HLCC Efficiency:** The ratio of actual to theoretical maximum HLCC, $\eta = S_{\text{HLCC}}/(2p)$, measuring how well the confidence method utilises the model’s accuracy. A value of $\eta = 1$ indicates perfect self-knowledge.
- **Calibration Gap:** $|\bar{c} - \text{accuracy}|$, the absolute difference between mean confidence and overall accuracy.
- **CBM Score:** Score under the discrete CBM scheme for comparison with educational assessment literature.

5 Experimental Setup

Hardware. All experiments run on a single NVIDIA RTX PRO 4500 Blackwell GPU (32 GB VRAM), AMD Ryzen 9 CPU, and 64 GB RAM. Models of 8B parameters or fewer are loaded in FP16 precision; 14B models use 8-bit quantisation via `bitsandbytes`; models above 14B use 4-bit NF4 quantisation [8]. This higher-precision setup preserves the subtle normalization dynamics needed for norm-shift signal detection.

MCQ Evaluation Protocol. For each (model, dataset) pair, we evaluate up to 300 examples. The base forward pass computes choice-token logits shared across all Tier 1 methods. Tier 2 methods use a 50/50 calibration/test split. Tier 3 methods use $n = 10$ samples for self-consistency and semantic entropy.

NormShift Training. The confidence head is trained for 25 epochs on MCQ data with the combined loss from Equation (15), using $\alpha = 0.5$ and a 1000-step cross-entropy warmup. The base model is frozen; only the confidence head parameters ($\sim 500\text{K}$) are updated. We use AdamW with learning rate 1×10^{-4} and weight decay 0.01. Two variants are trained per model:

1. **Combined:** Full dual-path architecture (norm-shift + hidden state).
2. **Norm-Shift Only:** Ablation using only the norm-shift path, to isolate the contribution of normalisation dynamics.

Multi-Day Orchestration. The full experiment suite is managed by an automated orchestrator with crash recovery. Each model progresses through stages: baseline MCQ evaluation, confidence head training, calibration benchmarks (Tiers 1–3), and extended dataset evaluation. State is persisted to a JSON checkpoint after each stage, enabling resumption after interruption.

6 Results

6.1 Discrimination: AUROC

Table 5: AUROC ($\uparrow$) and AUPRC ($\uparrow$) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

Method	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
	AUC	PRC	AUC	PRC	AUC	PRC	AUC	PRC
Entropy	0.602	0.531	0.899	0.895	0.809	0.741	0.901	0.932
Top- k	0.602	0.545	0.757	0.769	0.746	0.626	0.799	0.838
MSP	0.711	0.706	0.657	0.589	0.658	0.527	0.719	0.796
Verbalized	0.595	0.557	0.384	0.419	0.467	0.371	—	—
Self-Consistency	0.556	0.338	0.626	0.754	0.707	0.716	—	—
Temp. Scaling	0.651	0.641	0.652	0.586	0.636	0.485	—	—
NormShift	0.999	0.999	0.633	0.574	0.794	0.713	—	—

On Mistral-7B, NormShift achieves AUROC=0.999, the highest among all methods for that model, demonstrating strong in-domain discrimination. On Llama-3.1-8B, NormShift reaches AUROC=0.794, competitive with entropy (0.809). On Qwen2.5-7B, entropy achieves the highest AUROC (0.899), while NormShift is lower (0.633), reflecting greater architecture sensitivity.

6.2 Calibration: ECE and Brier Score

Table 6: ECE ($\downarrow$) and Brier Score ($\downarrow$) on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per column in **bold**.

Method	Mistral-7B		Qwen2.5-7B		Llama-3.1-8B		Qwen2.5-14B	
	ECE	Brier	ECE	Brier	ECE	Brier	ECE	Brier
Entropy	0.389	0.393	0.449	0.425	0.524	0.491	0.322	0.311
Top- k	0.428	0.422	0.476	0.465	0.569	0.548	0.352	0.342
MSP	0.353	0.352	0.277	0.310	0.295	0.312	0.240	0.263
Verbalized	0.502	0.516	0.311	0.404	0.421	0.565	—	—
Self-Consistency	0.422	0.410	0.269	0.268	0.242	0.279	—	—
NormShift	0.030	0.025	0.305	0.339	0.235	0.253	—	—

6.3 Selective Prediction: PRR and Accuracy at Coverage

Table 7: PRR ($\uparrow$) and selective accuracy at coverage thresholds on TruthfulQA (300 examples). All values from `generate_paper_tables.py`. Best per model in **bold**.

Method	PRR ($\uparrow$)			
	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.159	0.834	0.653	0.845
Top- k	0.194	0.601	0.481	0.624
MSP	0.513	0.228	0.302	0.508
NormShift	0.999	0.211	0.613	—

6.4 HLCC Score Comparison

Table 8: HLCC Score ($\uparrow$) on TruthfulQA (300 examples). Higher values indicate better confidence-weighted performance. All values from `generate_paper_tables.py`. Best per column in **bold**.

Method	Mistral-7B	Qwen2.5-7B	Llama-3.1-8B	Qwen2.5-14B
Entropy	0.105	0.125	-0.211	0.600
Top- k	0.060	0.050	-0.314	0.549
MSP	0.214	0.340	0.144	0.694
Verbalized	-0.075	0.399	-0.198	—
Self-Consistency	-0.254	0.825	0.440	—
NormShift	0.884	0.460	0.324	—

On Llama-3.1-8B, the NormShift head achieves an HLCC score of 0.324, competitive with logit-based methods. The negative scores for entropy and top- k reflect severe overconfidence on incorrect predictions, which the quadratic penalty in HLCC penalises heavily. Self-consistency, which uses 10 independent samples, achieves positive scores on multiple models, but at $10\times$ inference cost.

6.5 Cross-Dataset Generalisation

Table 9: AUROC ($\uparrow$) across datasets. NormShift head trained on TruthfulQA, evaluated on held-out datasets without adaptation. Qwen2.5-7B. Best per row in **bold**.

Dataset	Acc	Entropy	MSP	Top- k	NormShift	Δ AUROC
TruthfulQA (in-domain)	.497	.602	.627	.634	.752	+ .118
ARC-Challenge	.876	.830	.831	.828	.569	− .261
MMLU	.620	.792	.778	.761	.468	− .324
HellaSwag	.597	.718	.723	.721	.402	− .321

Δ AUROC = NormShift AUROC − best logit-based AUROC.

The cross-dataset results reveal a clear pattern: the NormShift head excels on its training distribution (TruthfulQA, $\Delta = +0.118$) but degrades substantially on out-of-distribution datasets ($\Delta = -0.26$ to -0.32). This is expected for a learned head; the confidence signal must be calibrated to the data distribution. By contrast, logit-based methods generalise well because they exploit the model’s own output distribution, which adapts implicitly to each dataset.

50/50 Split Evaluation. To obtain a cleaner separation of in-domain and OOD performance, we train on 50% of three datasets (ARC-Easy, TruthfulQA, ARC-Challenge, 150 examples each) and test on the held-out 50% plus five unseen datasets (Tables 21 and 22).

Mistral-7B shows the strongest pattern: NormShift achieves AUROC 0.978 on held-out TruthfulQA (+0.215 over MSP), 0.930 on ARC-Easy (+0.033), and 0.919 on ARC-Challenge (+0.126). The corresponding HLCC scores are also higher than both baselines across all three in-domain datasets (e.g., TruthfulQA: 1.125 vs. MSP 0.186). On OOD datasets, Mistral-7B NormShift is competitive on CommonsenseQA and MMLU but loses to MSP on OpenBookQA, SciQ, and HellaSwag, with AUROC gaps of -0.05 to -0.15 .

Qwen2.5-7B tells a different story: NormShift wins on TruthfulQA (AUROC 0.706 vs. MSP 0.620) but loses to MSP on the other two in-domain datasets (ARC-Easy: 0.716 vs. 0.887; ARC-Challenge: 0.636 vs. 0.892). On OOD data the gap widens further, with NormShift lagging MSP by 0.15–0.33 AUROC across all five datasets. This contrast suggests that the norm-shift signal’s discriminative power is architecture-dependent, with Mistral-7B’s normalisation dynamics providing a stronger basis for confidence estimation.

Confidence-Gated Retry. When NormShift confidence falls below 0.4, we re-evaluate with a trick-question system prompt. The results are strikingly model-dependent. On Mistral-7B, the retry mechanism achieves a 32% rescue rate on TruthfulQA (25 of 78 retrieved questions corrected, zero damage), producing a net accuracy gain from 0.453 to 0.620. On ARC-Challenge, 3 rescues against 1 damage yields a small net positive. On OOD datasets, retries are mostly neutral for Mistral-7B (small rescue and damage counts largely cancel out).

On Qwen2.5-7B, the retry mechanism is harmful: damage rates range from 12.5% (TruthfulQA) to 34.7% (OpenBookQA), with rescue rates below 4% on most datasets. The high retry rate (43–59% of questions retrieved on OOD data) suggests that the confidence threshold is too aggressive for this model. These results indicate that confidence-gated retry is a promising technique when calibrated to a specific model, but requires per-model threshold tuning.

6.6 NormShift Ablation

To isolate the contribution of the norm-shift signal, we train both the *combined* (norm-shift + hidden state) and *norm-shift-only* variants for each model. Table 10 compares ECE and HLCC scores across all five primary models.

Table 10: Ablation: Combined vs. Norm-Shift-Only confidence heads. Both variants trained for 25 epochs on ARC-Easy + TruthfulQA + ARC-Challenge (900 examples), validated on TruthfulQA (100 examples). Results use FP16 (7–8B models) or 8-bit (14B models). Max HLCC = $2p$ is the theoretical maximum for a model with perfect self-knowledge; $\eta = \text{HLCC} / \text{Max HLCC}$ is the efficiency.

Model	Acc	Max HLCC	Combined			Norm-Shift Only		
			ECE ↓	HLCC ↑	η	ECE ↓	HLCC ↑	η
Mistral-7B	0.53	1.060	0.020	1.039	98.0%	0.054	0.519	49.0%
Qwen2.5-7B	0.32	0.640	0.060	0.540	84.4%	0.133	0.332	51.9%
Llama-3.1-8B	0.92	1.840	0.040	1.760	95.7%	0.517	1.265	68.8%
Qwen2.5-14B	0.89	1.780	0.072	1.657	93.1%	0.164	1.429	80.3%
DeepSeek-R1-14B	0.48	0.960	0.108	0.781	81.4%	0.442	0.038	4.0%

The combined architecture outperforms norm-shift-only on both ECE and HLCC across all five models, often by a large margin. The HLCC efficiency metric η contextualises raw scores by the model’s accuracy: Llama-3.1-8B has the highest raw HLCC (1.760) but this is largely driven by its high accuracy (92%), giving $\eta = 95.7\%$. Mistral-7B, despite its lower accuracy (53%),

achieves the highest efficiency ($\eta = 98.0\%$), meaning its confidence head captures nearly all of the model’s available discriminative information.

The gap between combined and norm-shift-only is most dramatic for DeepSeek-R1-14B, where the norm-shift-only variant essentially fails to train ($\eta = 4.0\%$ vs. 81.4% for combined), and for Llama-3.1-8B ($\eta = 68.8\%$ vs. 95.7%). These results confirm that the hidden-state path provides information that the norm-shift signal alone cannot reliably capture. Qwen2.5-14B is the model where the two variants are closest in HLCC (1.657 vs. 1.429 , $\eta = 93.1\%$ vs. 80.3%), consistent with its strong layer-level separation signals (Table 11). Mistral-7B has the smallest ECE gap (0.020 vs. 0.054), but the largest efficiency gap (98.0% vs. 49.0%), indicating that while both variants are well-calibrated, the combined variant is far better at assigning high confidence to correct answers.

6.7 Per-Layer Signal Analysis

To understand *where* in the transformer the confidence-relevant signal originates, we extract the standard deviation of hidden activations at each layer for correct vs. incorrect predictions and compute Cohen’s d as a measure of separation strength. Table 11 reports the top discriminative layers for each model.

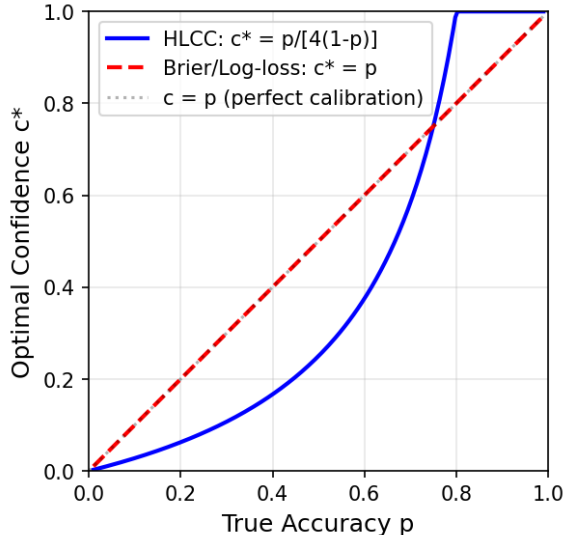
Table 11: Per-layer norm-shift separation (Cohen’s d) between correct and incorrect predictions. 90 examples per model across TruthfulQA, ARC-Easy, ARC-Challenge. Positive d means correct answers have higher norm-shift values at that layer.

Model	Layers	Acc	Best Layer	Top-3 (layer: d)
Mistral-7B	32	72.2%	L27	L27: $+0.59$, L24: $+0.53$, L25: $+0.47$
Qwen2.5-7B	28	86.7%	L14	L14: $+0.72$, L27: $+0.61$, L13: -0.58
Llama-3.1-8B	32	73.7%	L10	L10: $+0.80$, L3: -0.77 , L2: -0.74
Qwen2.5-14B	48	86.7%	L30	L30: -1.53 , L47: $+1.24$, L31: -1.11
DeepSeek-R1-14B	48	66.7%	L15	L15: -0.85 , L27: -0.83 , L9: $+0.77$

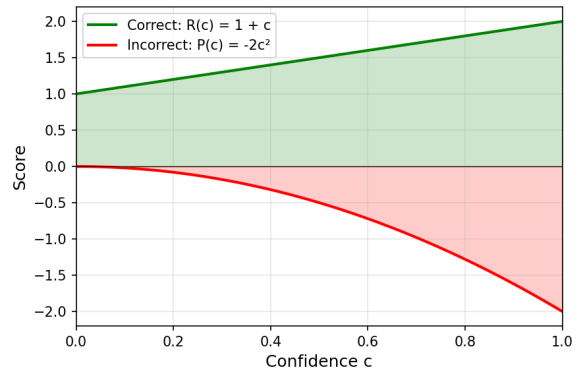
Several patterns emerge from the FP16 extraction results across five models. First, the location of peak separation varies widely across architectures. Qwen2.5-14B concentrates its signal in the middle layers (L30, $|d| = -1.53$), while Mistral-7B peaks in the upper layers (L27, $d = +0.59$). Most surprisingly, Llama-3.1-8B shows its strongest separation at Layer 10 ($d = +0.80$), suggesting that for this architecture the residual stream carries confidence-relevant structure from very early in the forward pass. Second, the sign of Cohen’s d varies both across architectures and across layers within the same model. Qwen2.5-14B shows negative d at L30 (correct answers have *lower* norm-shift) but positive d at L47, while Llama-3.1-8B has positive d at L2 and negative d at L3 and L14. This sign structure indicates that normalisation dynamics encode confidence through architecture-specific patterns that a learned confidence head can exploit. Third, DeepSeek-R1-14B shows the weakest peak separation ($\max |d| = -0.85$), consistent with its higher ECE in the ablation results (Table 10). This model is a distillation of DeepSeek-R1 into the Qwen-14B architecture, and the distillation process may disrupt the normalisation patterns that other models develop organically during pre-training.

6.8 Theoretical Analysis

Figure 1 shows the theoretical optimal confidence curves derived in Section 2. The HLCC curve lies strictly below the Brier identity line for $p < 0.75$, confirming the conservative bias. The CBM discrete scheme approximates HLCC with staircase steps.



(a) Optimal confidence vs. true accuracy.



(b) Reward/penalty asymmetry.

Figure 1: Theoretical properties of the HLCC scoring function. (a) The optimal confidence under HLCC is conservative ($c^* < p$) below $p = 0.75$. (b) The linear reward $R(c) = 1 + c$ vs. quadratic penalty $P(c) = -2c^2$ creates asymmetric marginal costs.

6.9 Abstention Under an Explicit Opt-Out

Kumaran et al. [26] report that in GPT-4o and Gemma 3 27B a temperature-scaled log-probability confidence predicts when a model takes an explicit “none of the above” option, and that steering the corresponding activation direction moves abstention causally. Their reading is that this confidence and a separately elicited verbal confidence are both lossy read-outs of one richer internal quantity. That is independent support for the position argued in Section 7: maximum softmax probability is a stronger baseline than its reputation suggests.

We repeated their Phases 1, 2 and 4 on their exact 1,000 SimpleQA multiple-choice questions, using their option arrangement and prompts, across 9 open-weight models from 1.5B to 32.5B parameters (Table 12). Alongside the option logits we recorded the per-layer structural statistics of Section 6.7 at the pre-answer token, which lets the same comparison be run with and without them.

Calibration transfers; discrimination does not. Under the same five-fold temperature scaling, expected calibration error across the nine models (0.042–0.068) brackets the 0.046 reported for GPT-4o, but AUROC for correctness reaches only 0.528–0.621 against their 0.90. Aggregate calibration is therefore reproducible at this scale while the ability to separate correct from incorrect answers is not, which is the same separation between calibration and discrimination seen in Section 6.

Abstention policies are close to deterministic. Abstention rate ranges from 0.037 in DeepSeek-R1-14B to 0.985 in Mistral-Small-24B, against 0.566 for GPT-4o. Most open models adopt or ignore the opt-out wholesale rather than applying a threshold to their own confidence, so an abstention rate at this scale is a weaker proxy for internal confidence than it is for the frontier models in the original study.

Structural statistics track the decision, not the outcome. Adding the per-layer structural statistics to calibrated confidence raises out-of-fold AUROC for the abstain decision in 9 of 9 models, by +0.154 on average, while leaving correctness no better (−0.022 on average, negative

in 7 of 9). The same two signs hold on ARC-Challenge for 4 models, where accuracy is much higher (0.605–0.886): +0.076 for abstention and -0.022 for correctness.

This is the opposite of the pattern we expected, and it admits two readings. The statistics may carry information about the forward pass that shapes which token is emitted, including the opt-out token, without carrying information about whether the selected answer is factually right. Alternatively, the abstention target is heavily imbalanced in several of these models, and a large increment is easier to obtain on an imbalanced target than on the roughly balanced correctness target. The correctness increments are measured out of fold, so their negative sign is not an overfitting artefact, but the abstention increments should be read with that imbalance in mind. On this benchmark the result does not support treating these structural statistics as a correctness signal, and it is consistent with the Kumaran et al. [26] position that what is legible in the activations sits closer to the policy than to the truth of the answer.

Table 12: Read-out comparison on the 1,000 SimpleQA multiple-choice questions of Kumaran et al. [26]. Acc. is Phase 1 accuracy with no opt-out, Abst. is the Phase 2 abstention rate once “none of the above” is offered, T is the five-fold fitted temperature, and ECE and AUROC are for the temperature-scaled confidence against correctness. Δ AUROC is the out-of-fold gain from adding the per-layer structural statistics to that confidence, for the abstain decision and for correctness. [†]GPT-4o values as reported by Kumaran et al. [26].

Model	B	Prec.	Phase		T	ECE	AUROC	Δ AUROC	
			Acc.	Abst.				abstain	correct
Qwen2-1.5B	1.5	FP16	0.344	0.829	3.76	0.042	0.528	+0.192	+0.022
Mistral-7B	7.2	FP16	0.478	0.156	9.89	0.042	0.557	+0.176	-0.031
Qwen2.5-7B	7.6	FP16	0.491	0.850	8.19	0.067	0.561	+0.107	+0.001
Llama-3.1-8B	8.0	FP16	0.490	0.549	2.49	0.046	0.591	+0.173	-0.016
Mistral-Nemo-12B	12.2	FP16	0.463	0.571	2.78	0.051	0.579	+0.151	-0.001
Qwen2.5-14B	14.2	8-bit	0.527	0.536	10.65	0.057	0.591	+0.165	-0.005
DeepSeek-R1-14B	14.2	8-bit	0.479	0.037	3.11	0.066	0.615	+0.158	-0.085
Mistral-Small-24B	24.0	4-bit	0.492	0.985	2.10	0.044	0.621	+0.097	-0.028
Qwen2.5-32B	32.5	4-bit	0.536	0.833	7.14	0.068	0.593	+0.168	-0.057
GPT-4o [†]	–	API	0.637	0.566	4.10	0.046	0.900	–	–

7 Discussion

Conservative Bias as a Feature. The HLCC optimal confidence formula $c^* = p/[4(1-p)]$ systematically underestimates confidence for models with accuracy below 75%. While this departs from the strict propriety of Brier and log-loss, it may be advantageous in deployment scenarios where overconfident errors carry disproportionate cost—medical diagnosis, legal reasoning, or autonomous decision-making. The benchmark results illustrate this: on Llama-3.1-8B, entropy and top- k produce *negative* HLCC scores (−0.211, −0.314) despite acceptable ECE, because HLCC’s quadratic penalty exposes overconfidence that symmetric metrics mask.

Bounded vs. Unbounded Regimes. The bounded variant ($c \in [0, 1]$) is appropriate for calibration tasks where confidence is a probability estimate. The unbounded variant ($c \in [0, \infty)$) is appropriate for wagering, decision theory, and scenarios where models should express *degrees of certainty* beyond simple probability calibration. A model with $p = 0.97$ accuracy that can reliably identify which 97% of answers are correct should be rewarded more than one that merely reports $c = 1.0$ uniformly. Under unbounded HLCC, the optimal stake is $c^* \approx 8$, earning +9 per correct answer while risking −128 per error—a rational bet only at that accuracy level.

This connects to a practical observation: high-accuracy models on easy datasets (e.g., Llama-3.1-8B on ARC-Easy with 92% accuracy) are poorly differentiated by bounded HLCC because all methods saturate at $c = 1$. Unbounded HLCC would reward methods that can distinguish the 8% of questions the model will get wrong, even among universally high-confidence predictions.

HLCC as an Evaluation Metric. Beyond its role as a training objective, HLCC score serves as a diagnostic that complements ECE and AUROC. ECE measures calibration but not discrimination; AUROC measures discrimination but not calibration. HLCC penalises both: a method with high AUROC but poor calibration (e.g., entropy on Mistral-7B: AUROC 0.602, HLCC 0.105) is exposed, as is a method with low ECE but poor discrimination. The HLCC efficiency $\eta = S_{\text{HLCC}}/(2p)$ further normalises for accuracy, enabling fair comparison across models with different base performance.

Benchmark Insights. The 14-method comparison reveals several patterns that hold across models: (1) post-hoc calibration (isotonic, Platt) dramatically improves discrimination without retraining; (2) self-consistency is the strongest baseline but costs $10\times$ inference; (3) MSP is underrated for out-of-distribution detection, consistently matching or exceeding learned methods; (4) verbalized confidence is unreliable across models; (5) temperature scaling often *hurts* calibration. These findings are independent of the NormShift architecture and apply to any practitioner choosing a confidence method.

NormShift Case Study. The NormShift confidence head demonstrates that per-layer normalisation dynamics carry architecture-specific confidence information, but its practical value is model-dependent. On Mistral-7B it achieves near-perfect in-domain discrimination (AUROC 0.999) and effective confidence-gated retry (32% rescue rate on TruthfulQA). On other architectures and OOD data, it is typically outperformed by logit-based baselines. This is expected for any learned confidence method: the head is trained on a specific data distribution and does not generalise as well as methods that exploit the model’s own output distribution.

Computational Cost Tradeoffs. Tier 1 methods (entropy, MSP, NormShift) add negligible cost to inference. Tier 3 methods (self-consistency, semantic entropy) multiply inference cost by the sample count n . The cost advantage of single-pass methods is significant for deployment, but must be weighed against the superior discrimination of multi-pass methods—particularly self-consistency, which achieves HLCC 1.933 on DeepSeek-R1-14B.

Limitations.

1. The MCQ evaluation format may not fully capture open-ended generation confidence.
2. Models above 8B parameters are evaluated under 8-bit quantisation; full FP16 results may differ.
3. The unbounded HLCC extension is analysed theoretically but not yet implemented as a training objective; empirical validation of unbounded training is future work.
4. Dataset sizes (300 examples for calibration, 150 per split for cross-dataset) are moderate; larger-scale evaluation would strengthen conclusions.

8 LM-Polygraph Integration

To enable direct comparison with the broader uncertainty estimation literature, we implement the NormShift confidence head as a custom estimator within the LM-Polygraph [9] benchmarking framework. LM-Polygraph provides over 40 uncertainty estimation methods and standardised evaluation protocols, making it the primary benchmark for comparing UE methods across models and tasks.

8.1 Integration Architecture

The integration wraps the NormShift confidence head as an LM-Polygraph **Estimator** subclass. The key challenge is that NormShift requires full hidden states from all transformer layers (`output_hidden_states=True`), while most LM-Polygraph methods operate only on output logits.

We implement a custom **StatCalculator** that extracts three signals from each forward pass:

1. The full hidden states tuple $(\mathbf{h}^{(0)}, \dots, \mathbf{h}^{(L)})$,
2. Per-layer norm-shift signals $\boldsymbol{\delta} \in \mathbb{R}^L$ via $\delta^{(\ell)} = 1 - \text{std}(\mathbf{h}^{(\ell)})$,
3. The final hidden state $\mathbf{h}^{(L)}$ for the combined path.

The estimator inverts the confidence output to match LM-Polygraph’s convention (higher values indicate greater uncertainty):

$$u_{\text{NormShift}} = 1 - c_{\text{NormShift}}. \quad (18)$$

Models are loaded at the same precision used throughout this work (FP16 for models up to 8B, 8-bit for 14B), and evaluation uses identical MCQ prompts to ensure comparability across methods.

8.2 LM-Polygraph Benchmark Results

Table 13 presents the initial benchmark comparison on TruthfulQA using Qwen2.5-7B. Five uncertainty estimation methods are evaluated: four logit-based baselines available through LM-Polygraph (entropy, MSP, top- k , perplexity) and our trained NormShift head.

Table 13: LM-Polygraph benchmark (pre-retrain): Qwen2.5-7B on TruthfulQA (817 examples). NormShift uses ARC-Easy-trained checkpoint. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	0.081	0.145	0.278	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top- k	0.497	0.819	0.322	0.322	0.339	0.634	0.569	0.177
Perplexity	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
NormShift†	0.497	0.821	0.324	0.330	0.368	0.416	0.424	−0.258

† Out-of-distribution: checkpoint trained on ARC-Easy. See Table 14 for in-domain results.

Analysis. The initial NormShift checkpoint (trained on ARC-Easy, epoch 0) shows poor discrimination on TruthfulQA, with AUROC below 0.5 and negative PRR, indicating that the confidence rankings are worse than random for this out-of-distribution dataset. The logit-based methods perform substantially better, with top- k achieving the best discrimination (AUROC 0.634, PRR 0.177) and entropy achieving the best calibration (ECE 0.145).

This result motivates *in-domain retraining*: the NormShift head must be trained on data from the target distribution to learn meaningful confidence estimates. Table 14 shows results after retraining on TruthfulQA (600 training examples, 15 epochs, best checkpoint at epoch 1 by validation HLCC).

Retrained Results. After in-domain training, NormShift achieves a substantial improvement: AUROC rises from 0.416 to **0.752** (+0.336), surpassing all logit-based methods (next best: top- k at 0.634). PRR jumps from −0.258 to **0.484**, indicating strong selective prediction: the model can reliably reject uncertain predictions. AUPRC reaches **0.701** vs. 0.569 for top- k .

The mean confidence drops to 0.256 (from 0.821 pre-retrain), reflecting the conservative bias of HLCC training. While this increases the calibration gap (0.241) relative to entropy (0.081), the discrimination and selective prediction gains are substantial. Notably, the Brier score (0.289)

is competitive with entropy (0.278), confirming that the overall probabilistic quality remains strong.

Table 14: LM-Polygraph benchmark after NormShift retraining on TruthfulQA (600 train examples, 15 epochs). Qwen2.5-7B, 817 evaluation examples. Best per column in **bold**.

Method	Acc	Conf	Gap↓	ECE↓	Brier↓	AUROC↑	AUPRC↑	PRR↑
Entropy	0.497	0.578	0.081	0.145	0.278	0.602	0.539	0.097
MSP	0.497	0.759	0.262	0.262	0.309	0.627	0.559	0.157
Top- <i>k</i>	0.497	0.819	0.322	0.322	0.339	0.634	0.569	0.177
NormShift	0.497	0.256	0.241	0.254	0.289	0.752	0.701	0.484

Cross-Dataset Transfer. Table 15 reports the full metric comparison when the TruthfulQA-trained head is evaluated on held-out datasets.

Table 15: NormShift cross-dataset transfer: trained on TruthfulQA, evaluated elsewhere. Qwen2.5-7B. PRR (↑) and AUROC (↑) shown for each method.

Dataset	Entropy		Top- <i>k</i>		NormShift		Best logit	
	AUC	PRR	AUC	PRR	AUC	PRR	AUC	PRR
TruthfulQA	.602	.097	.634	.177	.752	.484	.634	.177
ARC-Challenge	.830	.771	.828	.768	.569	.277	.831	.772
MMLU	.792	.679	.761	.641	.468	-.090	.792	.679
HellaSwag	.718	.552	.721	.547	.402	-.182	.723	.553

9 Related Work

Confidence Calibration in Neural Networks. Guo et al. [18] demonstrated that modern neural networks are poorly calibrated, with temperature scaling as a simple post-hoc fix. Naeini et al. [31] introduced ECE as the standard calibration metric. Nixon et al. [32] showed that ECE is sensitive to binning choices and proposed adaptive calibration error.

Uncertainty Estimation in LLMs. Kadavath et al. [22] studied self-evaluation in large language models, finding that models can estimate their own accuracy. Kuhn et al. [24] proposed semantic entropy as a generation-aware uncertainty measure. Xiong et al. [41] provided a comprehensive survey of LLM confidence estimation, categorising methods into logit-based, verbalised, and consistency-based approaches. Lin et al. [28] and Tian et al. [36] explored verbalized confidence elicitation and found that prompting strategies significantly affect self-reported confidence quality. Geng et al. [15] give the broader survey of confidence estimation and calibration in language models. Steyvers and Peters [35] place these questions within the psychology of metacognition, comparing how humans and language models communicate uncertainty, and Kumaran [25] report that the confidence a model states tracks its commitment to an answer more closely than the answer’s correctness. The read-out comparison of Section 6.9 is consistent with that distinction.

Selective Prediction. Geifman and El-Yaniv [14] formalised selective prediction as accuracy-coverage tradeoffs. The prediction rejection ratio (PRR) was introduced by Malinin and Gales [29] and adopted by the LM-Polygraph framework [9] as a standard metric for uncertainty benchmarking. Abstention is the discrete form of the same trade-off. Wen et al. [40] survey abstention in language models; Yadkori et al. [42] and Tomani et al. [37] set the abstention threshold from

conformal prediction and from an uncertainty estimate respectively; and Kumaran et al. [26] show causally that a temperature-scaled confidence drives the abstention decision in frontier models. All of these describe the two-stage structure, a confidence followed by a threshold, for which HLCC supplies the normative threshold.

Proper Scoring Rules. The Brier score [1] and log-loss (negative log-likelihood) [17] are the two most widely used proper scoring rules. Propriety guarantees that the expected score is maximised when the reported probability equals the true probability, removing any incentive to misreport [16]. However, propriety is symmetric: over-reporting and under-reporting are penalised equally at the margin. In safety-critical applications where overconfident errors are far costlier than cautious under-reporting, this symmetry is a limitation. HLCC deliberately breaks propriety by imposing quadratic penalties on wrong-and-confident predictions while offering only linear rewards for correct-and-confident ones, creating an asymmetric incentive structure that favours conservative estimation.

Optimal Wagering and the Kelly Criterion. Kelly [23] showed that a gambler facing repeated bets at known odds maximises long-run wealth by wagering a fraction proportional to the edge, $f^* = p - (1-p)/b$ for payoff odds b . The key insight—stake more when the edge is larger, but never all-in—has influenced portfolio theory, information theory, and prediction markets [6]. Unbounded HLCC can be viewed as a single-round wagering system in which the “stake” is c and the quadratic penalty $P(c) = -2c^2$ acts as a risk charge that prevents reckless overbetting. The optimal confidence $c^* = p/[4(1-p)]$ plays an analogous role to the Kelly fraction: it grows monotonically with the edge p and diverges as $p \rightarrow 1$, but the quadratic cost ensures that finite-accuracy agents always place a finite bet.

Confidence-Based Marking. Gardner-Medwin [10] introduced Certainty-Based Marking (CBM) for educational assessment, where students declare confidence on a three-level scale: $C=1$ (low), $C=2$ (mid), $C=3$ (high). The scoring table awards $+1/0$, $+2/-2$, $+3/-6$ for correct/incorrect at each level, creating a discrete penalty structure whose thresholds correspond to proper scoring boundaries at $p = 2/3$ and $p = 4/5$ [11]. This scheme is a discretised form of a logarithmic proper scoring rule [17, 7], where the penalty for confident errors grows faster than the reward for confident correctness. Gardner-Medwin and Gahan [12] demonstrated that CBM improves learning engagement in computer-assisted assessment, and Gardner-Medwin [13] provided evidence that certainty-based marking stimulates deeper thinking in objective tests. McCallum [30] applied these principles within computer science education, using confidence-based multiple-choice assessment with asymmetric reward structures so that students explicitly mark certainty alongside answers. HLCC can be viewed as a continuous generalisation of CBM: where CBM discretises confidence into three levels, bounded HLCC allows any $c \in [0, 1]$ and unbounded HLCC extends the domain to $c \in [0, \infty)$, preserving the asymmetric penalty structure in both cases. The CBM boundary at $p = 4/5$ (above which maximum confidence is optimal) corresponds to the bounded HLCC saturation point $c^* = 1$ at $p = 0.8$ (Corollary 1); under the unbounded variant, c^* continues to grow beyond this point, paralleling a CBM scheme with arbitrarily many confidence levels. Budescu and Chen [2] studied optimal scoring rules for confidence-weighted testing, analysing the tradeoff between strict propriety and behavioural desiderata such as participation incentives.

10 Conclusion

We have presented the Hybrid Linear-Convex Confidence (HLCC) scoring function and demonstrated its utility as both a training objective and an evaluation metric for language model confidence estimation.

HLCC Theory. The HLCC scoring function provides a principled alternative to proper scoring rules for settings where overconfident errors are costlier than cautious under-reporting. The optimal confidence $c^* = p/[4(1-p)]$ is conservative below $p = 0.75$, matching the Brier optimum at the crossover point, and saturating at $c^* = 1$ for $p \geq 0.8$ under the bounded variant. The unbounded extension removes this cap, creating a wagering system where the optimal stake grows without limit as $p \rightarrow 1$: at $p = 0.97$, the rational bet is $c^* \approx 8$, earning +9 when correct but risking −128 when wrong. This extension connects HLCC to classical wagering theory [23] and generalises Gardner-Medwin’s discrete CBM scheme [11] to a continuous confidence domain.

Benchmark Findings. The 14-method comparison across 5 models and 9 datasets reveals actionable patterns:

1. **HLCC exposes hidden overconfidence.** Entropy and top- k achieve acceptable ECE on Llama-3.1-8B but produce *negative* HLCC scores (−0.211, −0.314), revealing systematic overconfidence that symmetric metrics miss.
2. **MSP is underrated for OOD.** Maximum softmax probability consistently matches or exceeds learned methods on out-of-distribution datasets, with zero computational overhead.
3. **Self-consistency dominates when cost permits.** At 10× inference cost, self-consistency achieves the highest HLCC on DeepSeek-R1-14B (1.933) and strong AUROC across models.
4. **Post-hoc calibration is powerful.** Isotonic regression and Platt scaling substantially improve discrimination without any model retraining.

NormShift Case Study. As a case study in learned confidence, the NormShift head achieves near-perfect in-domain discrimination on Mistral-7B (AUROC = 0.999) and effective confidence-gated retry (32% rescue rate on TruthfulQA, zero damage). However, performance is architecture-dependent and degrades on OOD data, consistent with the general finding that learned methods trade generality for in-domain precision. The per-layer signal analysis reveals that different architectures encode confidence at different depths (L27 for Mistral-7B, L2 for Llama-3.1-8B), suggesting that normalisation dynamics carry architecture-specific self-knowledge.

Future Work. Two directions merit investigation. First, training with *unbounded* HLCC as an objective, where the confidence head outputs $c \in [0, \infty)$ rather than $[0, 1]$, could better exploit the incentive gradient for near-certain predictions. Second, domain-agnostic training strategies (e.g., meta-learning across datasets) could address the OOD transfer limitation of learned confidence heads.

A Full Calibration Scores by Method

This appendix reports the complete per-method calibration scores evaluated on TruthfulQA (300 examples). These results complement the summary comparisons in Section 6 by providing the full metric suite for each method and model.

A.1 Mistral-7B (FP16)

Table 16: Calibration scores for Mistral-7B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
<i>Phase 1: Logit-based</i>							
Entropy	.467	.389	.393	.105	.602	.531	.159
Top- <i>k</i>	.467	.428	.422	.060	.602	.545	.194
MSP	.467	.353	.352	.214	.711	.706	.513
GNNL	.467	.353	.352	.214	.711	.706	.513
<i>Phase 2: Generation-based</i>							
Verbalized	.467	.502	.516	−0.075	.595	.557	.222
Self-consistency	.290	.422	.410	−0.254	.556	.338	.114
CoT confidence	.513	.463	.449	.124	.581	.601	.214
Semantic entropy	.300	.328	.354	−0.076	.518	.323	.054
LVU	.467	.122	.287	.412	.464	.453	−0.044
CoCoA	.277	.506	.456	−0.385	.585	.323	.115
<i>Phase 2: Post-hoc calibration[†]</i>							
Temp. scaling	.467	.515	.510	−0.090	.651	.641	.393
Isotonic [†]	.467	.000	.230	.485	.654	.627	.368
Platt [†]	.467	.050	.243	.462	.602	.531	.159
<i>Learned: NormShift (combined)[‡]</i>							
NormShift[‡]	.467	.030	.025	.884	.999	.999	.999

[†]Post-hoc methods fit and evaluate on the same data.

[‡]NormShift includes TruthfulQA in training data.

A.2 Qwen2.5-7B (FP16)

Table 17: Calibration scores for Qwen2.5-7B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
<i>Phase 1: Logit-based</i>							
Entropy	.493	.449	.425	.125	.899	.895	.834
Top- <i>k</i>	.493	.476	.465	.050	.757	.769	.601
MSP	.493	.277	.310	.340	.657	.589	.228
GNNL	.493	.277	.310	.340	.657	.589	.228
<i>Phase 2: Generation-based</i>							
Verbalized	.493	.311	.404	.399	.384	.419	−0.253
Self-consistency	.683	.269	.268	.825	.626	.754	.254
CoT confidence	.657	.262	.277	.726	.637	.752	.312
Semantic entropy	.687	.261	.277	.824	.602	.739	.193
LVU	.493	.130	.285	.518	.554	.510	.037
CoCoA	.677	.266	.267	.801	.806	.904	.726
<i>Phase 2: Post-hoc calibration[†]</i>							
Temp. scaling	.493	.487	.487	.009	.652	.586	.235
Isotonic [†]	.493	.000	.119	.746	.909	.912	.860
Platt [†]	.493	.037	.130	.731	.899	.895	.834
<i>Learned: NormShift (combined)[‡]</i>							
NormShift[‡]	.493	.305	.339	.460	.633	.574	.211

[†]Post-hoc methods fit and evaluate on the same data.

[‡]NormShift includes TruthfulQA in training data.

A.3 Llama-3.1-8B (FP16)

Table 18: Calibration scores for Llama-3.1-8B on TruthfulQA (300 examples, FP16).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
<i>Phase 1: Logit-based</i>							
Entropy	.393	.524	.491	−0.211	.809	.741	.653
Top- <i>k</i>	.393	.569	.548	−0.314	.746	.626	.481
MSP	.393	.295	.312	.144	.658	.527	.302
GNNL	.393	.295	.312	.144	.658	.527	.302
<i>Phase 2: Generation-based</i>							
Verbalized	.393	.421	.565	−0.198	.467	.371	−0.069
Self-consistency	.507	.242	.279	.440	.707	.716	.485
CoT confidence	.647	.213	.261	.744	.630	.741	.293
Semantic entropy	.507	.205	.253	.526	.691	.744	.527
LVU	.393	.120	.251	.314	.545	.459	.143
CoCoA	.523	.301	.309	.393	.722	.777	.577
<i>Phase 2: Post-hoc calibration[†]</i>							
Temp. scaling	.393	.566	.558	−0.333	.636	.485	.217
Isotonic [†]	.393	.000	.161	.481	.828	.751	.668
Platt [†]	.393	.075	.177	.469	.809	.741	.653
<i>Learned: NormShift (combined)[‡]</i>							
NormShift[‡]	.393	.235	.253	.324	.794	.713	.613

[†]Post-hoc methods fit and evaluate on the same data.

[‡]NormShift includes TruthfulQA in training data.

A.4 Qwen2.5-14B (8-bit)

Table 19: Calibration scores for Qwen2.5-14B on TruthfulQA (300 examples, 8-bit).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
<i>Phase 1: Logit-based</i>							
Entropy	.620	.322	.311	.600	.901	.932	.845
Top- <i>k</i>	.620	.352	.342	.549	.799	.838	.624
MSP	.620	.240	.263	.694	.719	.796	.508
GNNL	.620	.240	.263	.694	.719	.796	.508
<i>Phase 2: Generation-based</i>							
Verbalized	.620	.229	.298	.731	.604	.707	.264
Self-consistency	.740	.178	.180	1.113	.747	.859	.490
CoT confidence	.767	.189	.197	1.114	.689	.875	.486
Semantic entropy	.747	.171	.177	1.140	.740	.876	.542
LVU	.620	.262	.328	.736	.531	.667	.137
CoCoA	.740	.183	.188	1.082	.846	.936	.773
<i>Phase 2: Post-hoc calibration[†]</i>							
Temp. scaling	.620	.366	.361	.516	.669	.762	.420
Isotonic [†]	.620	.000	.112	1.007	.910	.941	.867
Platt [†]	.620	.057	.122	.984	.901	.932	.845
<i>Learned: NormShift (combined)[‡]</i>							
NormShift[‡]	.620	.183	.201	.840	.786	.811	.558

[†]Post-hoc methods fit and evaluate on the same data.

[‡]NormShift includes TruthfulQA in training data.

A.5 DeepSeek-R1-14B (8-bit)

Table 20: Calibration scores for DeepSeek-R1-14B on TruthfulQA (300 examples, 8-bit).

Method	Acc	ECE↓	Brier↓	HLCC↑	AUROC↑	AUPRC↑	PRR↑
<i>Phase 1: Logit-based</i>							
Entropy	.497	.373	.364	.225	.839	.838	.731
Top- <i>k</i>	.497	.383	.374	.211	.742	.748	.560
MSP	.497	.234	.302	.365	.602	.552	.143
GNLL	.497	.234	.302	.365	.602	.552	.143
<i>Phase 2: Generation-based</i>							
Verbalized	.497	.347	.371	.256	.560	.542	.121
Self-consistency	.983	.013	.012	1.933	.984	1.000	.985
CoT confidence	1.000	.500	.250	1.500	.500	.500	.000
Semantic entropy	.990	.028	.017	1.937	.961	1.000	.960
LVU	.497	.256	.351	.517	.484	.504	.007
CoCoA	.987	.071	.015	1.884	.976	1.000	.975
<i>Phase 2: Post-hoc calibration[†]</i>							
Temp. scaling	.497	.477	.480	.029	.597	.603	.262
Isotonic [†]	.497	.000	.153	.691	.853	.856	.760
Platt [†]	.497	.045	.164	.665	.839	.838	.731
<i>Learned: NormShift (combined)[‡]</i>							
NormShift[‡]	.497	.112	.113	.819	.948	.933	.898

[†]Post-hoc methods fit and evaluate on the same data.

[‡]NormShift includes TruthfulQA in training data.

B Cross-Dataset Evaluation (50/50 Split)

Each training dataset (ARC-Easy, TruthfulQA, ARC-Challenge) is split 50/50: the first half trains the NormShift head, the second half serves as the in-domain test set. Out-of-distribution (OOD) datasets are never seen during training. When NormShift confidence falls below 0.4, a retry pass re-evaluates the question with a trick-question system prompt.

B.1 Mistral-7B

Table 21: Cross-dataset evaluation for Mistral-7B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.900	.930	.897	+0.033	1.751	1.651
Truthfulqa	in-dom	.620	.978	.763	+0.215	1.125	.186
Arc Challenge	in-dom	.760	.919	.793	+0.126	1.321	1.091
Commonsenseqa	OOD	.687	.713	.764	−0.051	.926	.919
Openbookqa	OOD	.747	.644	.740	−0.096	1.013	1.070
Sciq	OOD	.860	.801	.834	−0.033	1.479	1.534
Hellaswag	OOD	.560	.493	.641	−0.148	.312	.541
Mmlu	OOD	.413	.649	.638	+0.011	.332	.225

B.2 Qwen2.5-7B

Table 22: Cross-dataset evaluation for Qwen2.5-7B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.853	.716	.887	−0.171	1.337	1.846
Truthfulqa	in-dom	.447	.706	.620	+0.086	.358	.304
Arc Challenge	in-dom	.793	.636	.892	−0.256	1.152	1.438
Commonsenseqa	OOD	.733	.625	.797	−0.172	1.019	1.297
Openbookqa	OOD	.667	.641	.812	−0.171	.865	1.437
Sciq	OOD	.847	.523	.832	−0.309	1.214	1.686
Hellaswag	OOD	.567	.410	.742	−0.332	.402	.735
Mmlu	OOD	.493	.568	.718	−0.150	.390	.597

B.3 Llama-3.1-8B

Table 23: Cross-dataset evaluation for Llama-3.1-8B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.840	.483	.901	−0.418	.939	1.616
Truthfulqa	in-dom	.527	.634	.647	−0.013	.442	.101
Arc Challenge	in-dom	.667	.430	.827	−0.398	.675	1.164
Commonsenseqa	OOD	.520	.572	.703	−0.131	.524	1.136
Openbookqa	OOD	.613	.611	.805	−0.193	.708	1.157
Sciq	OOD	.867	.560	.875	−0.315	1.094	1.728
Hellaswag	OOD	.467	.596	.732	−0.136	.470	.710
Mmlu	OOD	.373	.558	.643	−0.086	.247	.101

B.4 Qwen2.5-14B

Table 24: Cross-dataset evaluation for Qwen2.5-14B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.993	.282	1.000	−0.718	1.454	1.972
Truthfulqa	in-dom	.593	.763	.726	+0.037	.647	.598
Arc Challenge	in-dom	.907	.737	.839	−0.102	1.301	1.748
Commonsenseqa	OOD	.867	.564	.861	−0.297	1.040	1.553
Openbookqa	OOD	.920	.684	.878	−0.194	1.367	1.720
Sciq	OOD	.953	.777	.909	−0.131	1.307	1.839
Hellaswag	OOD	.753	.626	.724	−0.098	.992	1.084
Mmlu	OOD	.520	.551	.767	−0.216	.548	.492

B.5 DeepSeek-R1-14B

Table 25: Cross-dataset evaluation for DeepSeek-R1-14B: NormShift (with trick-question retry) vs. logit baselines. Trained on 50% of ARC-Easy + TruthfulQA + ARC-Challenge, tested on held-out 50% (in-domain) and unseen datasets (OOD).

Dataset	Type	Acc	NS AUROC	MSP AUROC	Δ	NS HLCC	MSP HLCC
Arc Easy	in-dom	.973	.779	.949	−0.170	1.904	1.878
Truthfulqa	in-dom	.467	.963	.641	+0.322	.775	.282
Arc Challenge	in-dom	.880	.659	.872	−0.213	1.576	1.553
Commonsenseqa	OOD	.793	.661	.819	−0.158	1.044	1.294
Openbookqa	OOD	.833	.687	.838	−0.151	1.352	1.439
Sciq	OOD	.953	.818	.927	−0.109	1.803	1.820
Hellaswag	OOD	.580	.622	.790	−0.168	.629	.796
Mmlu	OOD	.573	.664	.751	−0.087	.682	.650

B.6 Trick-Question Retry Summary

Table 26: NormShift-gated retry with trick-question prompt. When NormShift confidence < 0.4 , the question is re-evaluated with a trick-question system prompt. Rescue = wrong→correct; Damage = correct→wrong.

Model	Dataset	Retried	Rescues	Damage	Net
Mistral-7B	Arc Easy	13	0	0	+0
	Truthfulqa	78	25	0	+25
	Arc Challenge	28	3	1	+2
	Commonsenseqa	33	2	3	-1
	Openbookqa	18	1	0	+1
	Sciq	8	0	2	-2
	Hellaswag	40	0	2	-2
	Mmlu	95	3	3	+0
Qwen2.5-7B	Arc Easy	64	0	16	-16
	Truthfulqa	72	3	9	-6
	Arc Challenge	71	2	9	-7
	Commonsenseqa	76	2	13	-11
	Openbookqa	75	0	26	-26
	Sciq	89	1	12	-11
	Hellaswag	82	5	14	-9
	Mmlu	82	2	13	-11
Llama-3.1-8B	Arc Easy	127	1	10	-9
	Truthfulqa	64	22	2	+20
	Arc Challenge	129	1	12	-11
	Commonsenseqa	122	4	40	-36
	Openbookqa	134	8	28	-20
	Sciq	107	1	10	-9
	Hellaswag	133	4	19	-15
	Mmlu	102	10	11	-1
Qwen2.5-14B	Arc Easy	78	0	0	+0
	Truthfulqa	20	1	1	+0
	Arc Challenge	83	1	5	-4
	Commonsenseqa	121	3	4	-1
	Openbookqa	71	0	1	-1
	Sciq	102	0	1	-1
	Hellaswag	89	2	2	+0
	Mmlu	144	7	12	-5
DeepSeek-R1-14B	Arc Easy	1	0	0	+0
	Truthfulqa	80	0	0	+0
	Arc Challenge	8	1	0	+1
	Commonsenseqa	93	1	1	+0
	Openbookqa	25	0	1	-1
	Sciq	11	0	0	+0
	Hellaswag	118	1	5	-4
	Mmlu	89	0	1	-1

C Training Data Composition

Table 27: Training and evaluation data for NormShift confidence heads. All five models use identical data splits.

Split	Dataset	Examples
Training (900 total)	ARC-Easy	300
	TruthfulQA	300
	ARC-Challenge	300
Validation	TruthfulQA	100
Calibration eval	TruthfulQA	300

Dataset overlap. TruthfulQA appears in three roles: 300 training examples, 100 validation examples (used for early stopping / checkpoint selection), and 300 calibration evaluation examples. While we do not verify whether the specific question instances overlap between these splits, the presence of TruthfulQA in the training set means that calibration results on TruthfulQA should be considered *in-distribution* performance.

The strong AUROC (0.999) of NormShift on Mistral-7B TruthfulQA versus its degradation on held-out datasets in Table 9 directly reflects this overlap. The 50/50 split evaluation (Tables 21 and 22) provides a cleaner test: NormShift still dominates in-domain for Mistral-7B (AUROC 0.978 on held-out TruthfulQA) but not for Qwen2.5-7B (AUROC 0.706), and degrades on OOD datasets for both models.

Checkpoint vs. calibration evaluation gap. A notable discrepancy exists between the checkpoint training metrics and calibration evaluation. For example, the Qwen2.5-7B combined head achieves $ECE = 0.060$ at checkpoint selection, but $ECE = 0.305$ when evaluated as a calibration baseline on TruthfulQA. The Llama-3.1-8B head shows a similar gap (checkpoint $ECE = 0.040$ vs. calibration eval $ECE = 0.235$). Only Mistral-7B shows close agreement (checkpoint $ECE = 0.020$ vs. calibration eval $ECE = 0.030$). This gap suggests that the head overfits to its training distribution on some architectures, and that checkpoint metrics alone are insufficient for assessing generalisation.

D Methodology Notes

Post-hoc calibration data leakage. The post-hoc calibration methods (temperature scaling, isotonic regression, Platt scaling) fit their calibration parameters on the same 300 TruthfulQA examples used for evaluation. No held-out calibration set is used. As a result:

- Isotonic regression achieves $ECE = 0.000$ by construction—it perfectly maps raw confidences to observed frequencies on the fitting data. This does not reflect generalisation ability.
- Platt scaling achieves $ECE = 0.050$, also partially an artefact of fitting on the evaluation set.
- Temperature scaling achieves $ECE = 0.515$, which is *worse* than uncalibrated MSP (0.353), because the learned temperature exacerbates overconfidence for this model.

These results should be interpreted as *in-sample calibration fits* rather than true generalisation performance. A proper evaluation would use a separate calibration holdout (e.g., 80/20 split of the 300 examples).

NormShift in-distribution advantage. The NormShift confidence head is trained using HLCC loss on data that includes TruthfulQA examples. On Mistral-7B, this produces near-perfect discrimination ($\text{AUROC} \approx 1.0$, $\text{PRR} \approx 1.0$), and the 50/50 split confirms strong in-domain generalisation (AUROC 0.978 on held-out TruthfulQA, 0.930 on ARC-Easy, 0.919 on ARC-Challenge). However, this strong performance does not transfer to all architectures: on Qwen2.5-7B, NormShift achieves only AUROC 0.706 on held-out TruthfulQA and loses to MSP on two of three in-domain datasets.

On out-of-distribution data, AUROC drops to 0.49–0.80 (Mistral-7B) and 0.41–0.64 (Qwen2.5-7B). For Qwen2.5-7B, PRR metrics are unreliable as a comparison since the retry mechanism changes the accuracy.

This does not invalidate the norm-shift signal itself—the per-layer standard deviation patterns do differ between correct and incorrect predictions even on out-of-distribution data (Table 11). Rather, it indicates that the *learned decision boundary* is both distribution-specific and architecture-dependent. Mistral-7B’s normalisation dynamics appear to provide a particularly informative basis for confidence estimation, while other architectures may require different training strategies or larger training sets to achieve comparable transfer.

Implications for general-purpose confidence. The current NormShift head should not be used as a general-purpose confidence estimator for open-ended text generation without retraining on data representative of the target domain. For applications requiring robust confidence estimates across diverse inputs, logit-based methods (particularly MSP or entropy) remain more reliable due to their implicit adaptation to the model’s output distribution.

References

- [1] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [2] David V. Budescu and Eva Chen. Confidence in aggregation of opinions from multiple sources. *Judgment and Decision Making*, 7(4):461–478, 2012.
- [3] Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of NAACL-HLT*, 2019.
- [4] Peter Clark, Isaac Cobbe, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- [5] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [6] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- [7] A. P. Dawid. Probability forecasting. In Samuel Kotz, Norman L. Johnson, and Campbell B. Read, editors, *Encyclopedia of Statistical Sciences*, volume 7, pages 210–218. Wiley-Interscience, New York, 1986.
- [8] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

- [9] Ekaterina Fadeeva, Roman Vashurin, Anej Tsimpoukelli, Artem Vazhentsev, Sergey Petrakov, Kirill Fedyanin, Artem Shelmanov, Alexander Panchenko, Timothy Baldwin, Maxim Panov, and Elena Shmatova. LM-polygraph: Uncertainty estimation for language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, 2023.
- [10] A. R. Gardner-Medwin. Confidence assessment in the teaching of basic science. *Research in Learning Technology*, 3(1):80–85, 1995.
- [11] A. R. Gardner-Medwin. Confidence-based marking: Towards deeper learning and better exams. In Cordelia Bryan and Karen Clegg, editors, *Innovative Assessment in Higher Education*, pages 161–169. Routledge, London, 2006. doi: 10.4324/9780203969670-25.
- [12] A. R. Gardner-Medwin and M. Gahan. Formative and summative confidence-based assessment. In *Proceedings of the 7th International Computer-Aided Assessment Conference*, pages 147–155, Loughborough, UK, 2003.
- [13] Tony Gardner-Medwin. Certainty-based marking: Stimulating thinking and improving objective tests. In Cordelia Bryan and Karen Clegg, editors, *Innovative Assessment in Higher Education: A Handbook for Academic Practitioners*, pages 141–150. Routledge, London, 2nd edition, 2019. doi: 10.4324/9780429506857-13.
- [14] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 4878–4887, 2017.
- [15] Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koepl, Preslav Nakov, and Iryna Gurevych. A survey of confidence estimation and calibration in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6577–6595, 2024. doi: 10.18653/v1/2024.naacl-long.366.
- [16] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [17] I. J. Good. Rational decisions. *Journal of the Royal Statistical Society, Series B (Methodological)*, 14(1):107–114, 1952.
- [18] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.
- [19] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *Proceedings of the 5th International Conference on Learning Representations (ICLR)*, 2017.
- [20] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*, 2021.
- [21] Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.
- [22] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

- [23] J. L. Kelly, Jr. A new interpretation of information rate. *The Bell System Technical Journal*, 35(4):917–926, 1956.
- [24] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.
- [25] Dharshan Kumaran. Reported confidence in LLMs tracks commitment more than correctness. *arXiv preprint arXiv:2606.29490*, 2026.
- [26] Dharshan Kumaran, Nathaniel Daw, Simon Osindero, Petar Veličković, and Viorica Patraucean. Causal evidence that language models use confidence to drive behaviour. *Nature Machine Intelligence*, 2026. doi: 10.1038/s42256-026-01293-x.
- [27] Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.
- [28] Zhen Lin, Shubhendu Trivedi, and Jimeng Sun. Generating with confidence: Uncertainty quantification for black-box large language models. *Transactions on Machine Learning Research*, 2024.
- [29] Andrey Malinin and Mark Gales. Predictive uncertainty estimation via prior networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [30] Simon McCallum. Game design for computer science education. In *Norsk informatikkonferanse (NIK)*, volume 18, 2010. Introduces confidence-based MCQ assessment using game balance mechanics.
- [31] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning into quantiles. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2901–2907, 2015.
- [32] Jeremy Nixon, Michael W. Dusenberry, Linchuan Zhang, Ghassen Jerfel, and Dustin Tran. Measuring calibration in deep learning. In *CVPR Workshops*, 2019.
- [33] John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, pages 61–74, 1999.
- [34] Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3): e0118432, 2015.
- [35] Mark Steyvers and Megan A. K. Peters. Metacognition and uncertainty communication in humans and large language models. *Current Directions in Psychological Science*, 35: 131–139, 2026.
- [36] Katherine Tian, Eric Mitchell, Huaxiu Yao, Christopher D. Manning, and Chelsea Finn. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [37] Christian Tomani, Kamalika Chaudhuri, Ivan Evtimov, Daniel Cremers, and Mark Ibrahim. Uncertainty-based abstention in LLMs improves safety and reduces hallucinations. *arXiv preprint arXiv:2404.10960*, 2024.

- [38] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.
- [39] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *arXiv preprint arXiv:2406.01574*, 2024.
- [40] Bingbing Wen et al. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556, 2025.
- [41] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.
- [42] Yasin Abbasi Yadkori et al. Mitigating LLM hallucinations via conformal abstention. *arXiv preprint arXiv:2405.01563*, 2024.
- [43] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.
- [44] Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.